

ASSIGNMENT 2

• SESSION 13

1. Create a decision table for a Flipkart-style coupon validation system where a coupon is only valid if: (1) the user is logged in, (2) the cart value is above ₹500, and (3) the coupon has not expired. Show all possible combinations and the resulting action (accept/reject).

ANS.

1. Decision Table – Flipkart-style Coupon Validation :-

Conditions :-

- **C1:** User is logged in?
- **C2:** Cart value > ₹500?
- **C3:** Coupon has not expired

Action :-

- **Accept Coupon**
- **Reject Coupon**

Rule	User Logged In	Cart > ₹500	Coupon Not Expired	Action
R1	Yes	Yes	Yes	Accept
R2	Yes	Yes	No	Reject
R3	Yes	No	Yes	Reject
R4	Yes	No	No	Reject
R5	No	Yes	Yes	Reject
R6	No	Yes	No	Reject
R7	No	No	Yes	Reject
R8	No	No	No	Reject

Conclusion :-

The coupon is accepted **only when all three conditions are satisfied** (Rule R1). Any missing condition results in rejection.

2. Draw a state transition diagram for a Paytm login feature that locks the account after 3 failed password attempts and only unlocks after a successful OTP verification.

Ans.

Current State	Event	Next State
Logged Out	Correct password	Logged In
Logged Out	Wrong password (1st)	Failed Attempt 1
Failed Attempt 1	Wrong password	Failed Attempt 2
Failed Attempt 2	Wrong password	Account Locked
Account Locked	Correct OTP	Logged Out
Account Locked	Wrong OTP	Account Locked

3. Given the following use case: 'A Zomato user can place an order, cancel it before it is prepared, or track the order once prepared,' create a use case diagram and list at least 3 possible test scenarios based on it.

Ans.

Use Case Diagram

- User
 - (1) Place Order
 - (2) Cancel Order
 - (3) Track Order

Test Scenario	Expected Result
Place an order successfully	Order is placed successfully.
Cancel order before preparation	Order is cancelled successfully.
Try cancelling after preparation	Cancellation is not allowed.
Track order after preparation	Order tracking is available.
Track order before preparation	Appropriate status is shown.

4. Modify your decision table from Task 1 to include an additional rule: the coupon cannot be used more than twice by the same user. Update the table and explain how this new rule changes the possible outcomes.

Ans.

Updated Decision Table (Coupon Usage Limit Added) :-

Additional Condition :-

- **C4:** Coupon used **2 times or fewer** by the same user?

Conditions :-

- C1: User logged in?
- C2: Cart value > ₹500?
- C3: Coupon not expired?
- C4: Usage count ≤ 2?

Updated Decision Table :-

Rule	Logged In	Cart > ₹500	Not Expired	Usage ≤ 2	Action
R1	Yes	Yes	Yes	Yes	Accept
R2	Yes	Yes	Yes	No	Reject
R3	Yes	Yes	No	Yes	Reject
R4	Yes	Yes	No	No	Reject
R5	Yes	No	Yes	Yes	Reject
R6	Yes	No	Yes	No	Reject
R7	Yes	No	No	Yes	Reject

Effect of the New Rule :-

Previously, the coupon was accepted when:

- User was logged in,
- Cart value was greater than ₹500, and
- Coupon had not expired.

After adding the **usage count** condition, the coupon is accepted **only if all four conditions are true**, including that the coupon has been used **no more than twice** by the same user. If the user has already used the coupon more than two times, the system rejects it even if all the other conditions are satisfied.